

Adaptive Learning Rate

❑ Stochastic gradient descent to find a local optima

- ❑ Default choice for η : a fixed, small positive constant $\theta_{t+1} = \theta_t - \eta g_t$
- ❑ Problems: slow progress, or jump over 'gradient cliff', or oscillation

❑ Adaptive learning rate: let η change over epoch

- ❑ Strategy #1: $\eta_t = \frac{1}{t} \eta_0$
- ❑ Strategy #2: $\eta_t = (1 - \frac{t}{T})\eta_0 + \frac{t}{T}\eta_\infty$ if $t \leq T$ and $\eta_t = \eta_\infty$
 - ❑ Intuitions: smaller adjustment as algorithm progresses
 - ❑ e.g., $\eta_0 = 0.9$; $\eta_\infty = 10^{-9}$
- ❑ Strategy #3 (AdaGrad): $\eta_t = \frac{1}{\rho + r_i} \eta_0$ $r_i = \sqrt{\sum_{k=1}^{t-1} g_{i,k}^2}$
 - ❑ Intuition: The magnitude of gradient g_t : indicator of the overall progress
 - ❑ e.g., $\rho = 10^{-8}$
- ❑ Strategy #4 (RMSProp): exponential decaying weighted sum of squared historical gradients

